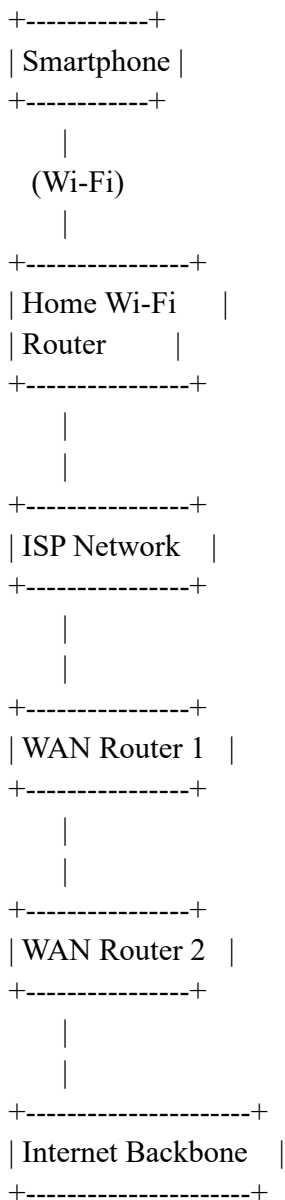


N+ - Network security, Maintenance and Troubleshooting procedures

Wide Area Network

[Q-1] Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home Wi-Fi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.

Aim: To show how a smartphone connects to Instagram servers through a home Wi-Fi network and the Internet. Network Diagram



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|
|
+-----+
| Instagram Servers |
+-----+

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Description

1. The smartphone connects to the home Wi-Fi router.
2. The router forwards data to the Internet Service Provider (ISP).
3. The data passes through multiple WAN routers on the Internet.
4. Finally, the request reaches Instagram's servers, and the response follows the same path back to the phone.

[Q-2] Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato offices, IRCTC ticketing, your home).

WAN Technology	Speed	Cost	Real-Life Example
MPLS (Multiprotocol Label Switching)	High	High	Large companies such as Zomato offices connecting branches
Leased Line	Very High	Very High	Banks, IRCTC ticketing systems, Data Centers
Broadband Internet	Medium to High	Low	Home Internet, Small Businesses, Schools

Conclusion

- MPLS is suitable for organizations requiring secure communication between multiple offices.
- Leased Lines provide dedicated, reliable, high-speed connectivity.
- Broadband Internet is the most affordable and commonly used WAN technology for homes.

[Q-3] Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop.
 Hint: Focus on how the VPN creates a private tunnel over the public WAN. What is a VPN?

A Virtual Private Network (VPN) creates a secure, encrypted connection (called a tunnel) between your device and a VPN server while using a public network such as the Internet. Example

Suppose you are connected to a public Wi-Fi at a coffee shop and want to access Spotify or YouTube.



Normally:

Your Device → Public Wi-Fi → Internet → Website

With VPN:

Your Device → Encrypted VPN Tunnel → VPN Server → Internet → Spotify/YouTube

How VPN Uses WAN

- The Internet is a Wide Area Network (WAN).
- A VPN creates an encrypted tunnel across the public WAN.
- Data travelling through the WAN cannot easily be read by hackers.
- The VPN server decrypts the data before forwarding it to the destination server.

Advantages

- Protects personal information.
- Encrypts Internet traffic.
- Prevents data theft on public Wi-Fi.
- Provides secure remote access to company networks.

[Q-4] Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

Scenario

Flipkart wants to connect warehouses located in:

- Ahmedabad
- Mumbai
- Delhi

Recommended WAN Technology MPLS (Multiprotocol Label Switching) Justification Reason 1: Reliable and Fast Communication

MPLS provides low latency, stable connectivity, and fast communication between all warehouses. This is important for inventory updates, order processing, and warehouse management.

Reason 2: Secure Private Network

MPLS offers a private network instead of using the public Internet. This improves security for sensitive business data and reduces the risk of cyber threats.